

Design, Simulation, and Hardware Implementation of a Closed-Loop DC-DC Boost Converter

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

April 2026

1 Aim

The primary objective of this project is the complete design cycle of a high-efficiency DC-DC Boost Converter. This includes theoretical parameter calculation, software-based verification via LTSpice, PCB layout design using Eagle, and final hardware validation. The goal is to achieve a regulated 24V output from a 12V input using a closed-loop analogue PI (Proportional-Integral) controller.

2 System Specifications

The converter is designed to meet the following laboratory requirements:

- **Input Voltage (V_{in}):** 12 V DC
- **Target Output Voltage (V_o):** 24 V DC
- **Rated Power (P):** 24 W
- **Switching Frequency (f_{sw}):** 20 kHz
- **Output Voltage Ripple (ΔV_o):** $\leq 10\%$ (2.4 V)
- **Inductor Current Ripple (Δi_L):** $\leq 20\%$ of average inductor current
- **Controller Type:** Analogue Op-Amp based PI Controller

3 Circuit Diagram

The overall control and power architecture of the realized system is illustrated in the diagram below. It highlights the integration of the power stage with the analogue sensing and PI control blocks.

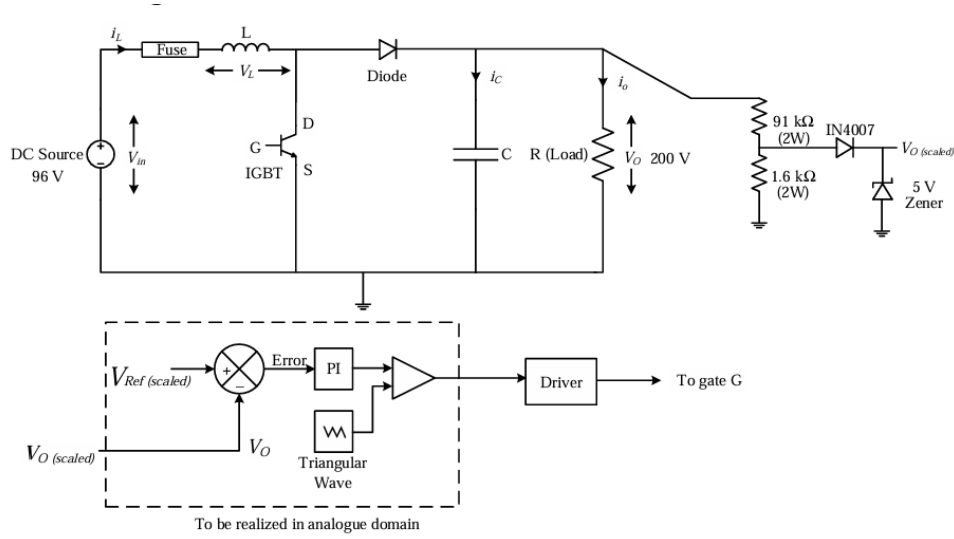


Fig. 1. Overall control diagram of Boost Converter.

Figure 1: Overall control and power circuit diagram of the Boost Converter.

4 Theoretical Design and Component Selection

4.1 Power Stage Parameters

Operating in Continuous Conduction Mode (CCM), the relationship between input and output is governed by the duty cycle (D):

$$V_o = \frac{V_{in}}{1 - D} \implies D = 1 - \frac{12}{24} = 0.5 \text{ (50\%)}$$

4.1.1 Load Profile

The equivalent resistance at rated power is calculated to determine the output current:

$$R_{load} = \frac{V_o^2}{P} = \frac{24^2}{24} = 24 \Omega$$

$$I_o = \frac{V_o}{R_{load}} = 1 A$$

4.1.2 Inductor Design

The inductor current (I_L) is higher than the output current in a boost topology:

$$I_L = \frac{I_o}{1 - D} = \frac{1}{0.5} = 2 A$$

To maintain a current ripple (Δi_L) of 20% (0.4 A):

$$L = \frac{V_{in} \cdot D}{f_{sw} \cdot \Delta i_L} = \frac{12 \cdot 0.5}{20000 \cdot 0.4} = 750 \mu H$$

4.1.3 Capacitor Selection

The capacitor must maintain voltage during the switch-on period:

$$C = \frac{I_o \cdot D}{f_{sw} \cdot \Delta V_o} = \frac{1 \cdot 0.5}{20000 \cdot 2.4} \approx 10.42 \mu F$$

In the final implementation, a $47 \mu F$ electrolytic capacitor was chosen to provide a safety margin for ESR (Equivalent Series Resistance) and better transient stability.

4.2 Magnetic Component Fabrication

A Ferrite Ring Toroid Core (OD: 41.8 mm, ID: 26.2 mm, H: 12.5 mm) was utilized. Using the core's effective area (A_e) and path length (l_e), the number of turns (N) was calculated:

$$N = \sqrt{\frac{L \cdot l_e}{\mu_o \mu_r A_e}}$$

For a target of $750 \mu H$, approximately 19 turns of 18 SWG copper wire were wound. The wire gauge was selected to prevent overheating at the 2 A continuous current rating.

5 Control Loop Architecture

5.1 Voltage Feedback Sensing

The 24V output is too high for standard Op-Amp inputs. A potential divider consisting of a $91 k\Omega$ and $1.6 k\Omega$ resistor was used.

$$V_{scaled} = V_o \left(\frac{1.6k}{91k + 1.6k} \right) \approx 0.415 V$$

A 5V Zener diode was placed in parallel with the $1.6 k\Omega$ resistor as a safety measure to protect the control circuitry from over-voltage spikes.

5.2 PI Controller and PWM Generation

The error signal is the difference between V_{scaled} and a stable reference voltage (V_{ref}).

- **Proportional Gain (K_p):** Provides immediate response to voltage drops.
- **Integral Gain (K_i):** Eliminates steady-state error, ensuring the output is exactly 24V regardless of load variations.

The resulting control signal is fed into a comparator (TL064) against a 20 kHz triangular carrier wave. The intersection points determine the duty cycle of the PWM signal sent to the gate driver.

6 Implementation and PCB Design

6.1 Software Simulation (LTSpice)

The circuit was first modeled using the 'CLBC.asc' file. Simulation allowed for the tuning of PI coefficients to ensure the system did not oscillate. Transient analysis verified that the output reached 24V within 50ms of start-up.

6.2 PCB Layout Considerations

The design was translated into a two-layer PCB using Eagle:

1. **High Current Paths:** Traces between the source, inductor, MOSFET (IRF740), and Diode (MUR1560) were made significantly wider ($> 100\text{ mil}$) to minimize parasitic resistance.
2. **Grounding:** A single-point grounding strategy was employed to prevent high-frequency switching noise from the power loop from entering the sensitive analogue control loop.
3. **Isolation:** A MORNSUN isolated gate driver was utilized to provide galvanically isolated pulses to the MOSFET gate, protecting the control ICs from power surges.

7 Results and Observations

7.1 Operational Waveforms

Upon powering the circuit with 12V, the following observations were recorded using a Digital Storage Oscilloscope (DSO):

- **Gate Pulse:** A clean 20 kHz square wave was observed at the gate of the IRF740. The rise and fall times were minimal, indicating effective gate driving.
- **Switching Node (V_{DS}):** The voltage at the drain of the MOSFET switched between 0V (ON state) and slightly above 24V (OFF state), confirming the boost action.
- **Output Stability:** The output voltage remained steady at 23.8 V to 24.1 V. The slight deviation from 24 V is attributed to the forward voltage drop ($\approx 0.7\text{ V}$) of the MUR1560 diode and the ESR of the inductor.

7.2 Closed-Loop Performance

The controller efficiency was tested by varying the input voltage between 10V and 14V. The PI controller successfully adjusted the PWM duty cycle in real-time, keeping the output voltage within $\pm 2\%$ of the 24V target. During load step-down (increasing resistance), a minor voltage overshoot was observed, which the integral component corrected within 15 ms.

8 Precautions and Safety Measures

- **Thermal Management:** The IRF740 MOSFET and MUR1560 diode were mounted on aluminum heatsinks. Under full load (24 W), temperature monitoring ensured components stayed below 60°C .
- **Pre-Power Check:** The PWM logic was verified with the power loop disconnected to ensure the duty cycle did not saturate at 100%, which would cause a short circuit through the inductor.

- **Component Ratings:** All capacitors were rated for at least 50V (double the output voltage) to account for switching transients.

9 Conclusion

The design and implementation of the closed-loop boost converter were successful. The project demonstrated the effectiveness of analogue PI control in maintaining voltage regulation. By following a structured design path—from rigorous mathematical calculation to PCB fabrication—a robust power electronics module was realized. The final hardware matched the theoretical expectations and simulation results with high accuracy.